TravelTide

FOR ELENA TARRANT, HEAD OF MARKETING

Which perk will make each customer join TravelTide Rewards?

ASK 1

Is there evidence for your five perks?

Free hotel meal · free checked bag · no cancellation fees · exclusive discounts · 1 night free hotel with flight

ASK 2

Which perk for each customer?

One likely favourite perk per customer, so the invitation email can lead with it

Jacklynn Tong · Data Analytics · TravelTide Mastery Project · 6 Nov 2026

WHAT WE DID

From 1 million users to one perk each

1,020,926
all TravelTide users

979,801
active since Jan 4, 2023

5,998 customers
with more than 7 sessions
83 % USA · median age 41
49,211 sessions · 15,489 kept trips

1. **Prepare**
   Cleaned the data and described each customer by party size, nights, distance, discounts, repeat cities
2. **Find groups**
   8 groups from clear habits (e.g. hotel-only stays) – each tested: real habit or just luck?
3. **Double-check**
   A clustering algorithm (K-Means), not told our rules, found 4 of the same groups on its own
4. **Assign perks**
   76 % get their group's perk; the 24 % with no clear habit get the perk they score highest on

Cohort as Elena suggested: sessions from Jan 4, 2023 · users with more than 7 sessions · SQL and Python notebooks on GitHub

The short answer

ASK 1 · EVIDENCE FOR YOUR FIVE PERKS?

Partly yes

- **Backed by a habit:** free hotel meal, no cancellation fees, 1 night free hotel with flight
- **No clear habit – test first:** exclusive discounts, free checked bag
- **New idea from the data:** an extra-legroom seat for long-haul travellers

ASK 2 · A PERK FOR EACH CUSTOMER?

Yes – all 5,998

- **9 customer groups** share 6 perks – one perk per customer
- **76 %** by a clear segment rule, **24 %** by a perk score
- Saved for every customer in **user_perks.csv**

Now live in Tableau

BACKUP · UNDER THE HOOD 1/3

What we fixed before counting anything

Issue	What we did	Effect
Cancellations	A cancelled trip is not travel: its booking doesn't count	610 trips out · 15,489 kept
Hotel nights	Raw nights had negative values → recalculated from check-in and check-out dates	421 trips with invalid dates left out of spend
Discount flags	Flags show what was offered → counted only on the part that was booked	549 trips corrected: 26.3 % → 22.7 % discounted
Birth year 2006	Placeholder birthdates: 233 users, almost all of the under-18s	Under-18s left out of the personas
Time window	7 cancelled trips were booked before Jan 4, 2023	Kept – no effect on trips

Details: notebook 01 (EDA & cleaning) and the SQL file on GitHub

BACKUP · UNDER THE HOOD 2/3

Three checks behind every perk

SHUFFLE TEST

50 ×

Trips dealt out again at random. 5 rules beat 95 % of the shuffles – habits. Bags, discounts and weekends don’t – luck.

CHANCE MODELS

458 vs 236

Customers who never book vs what chance predicts – the Dreamers are a real group, not unlucky browsers.

REDUNDANCY

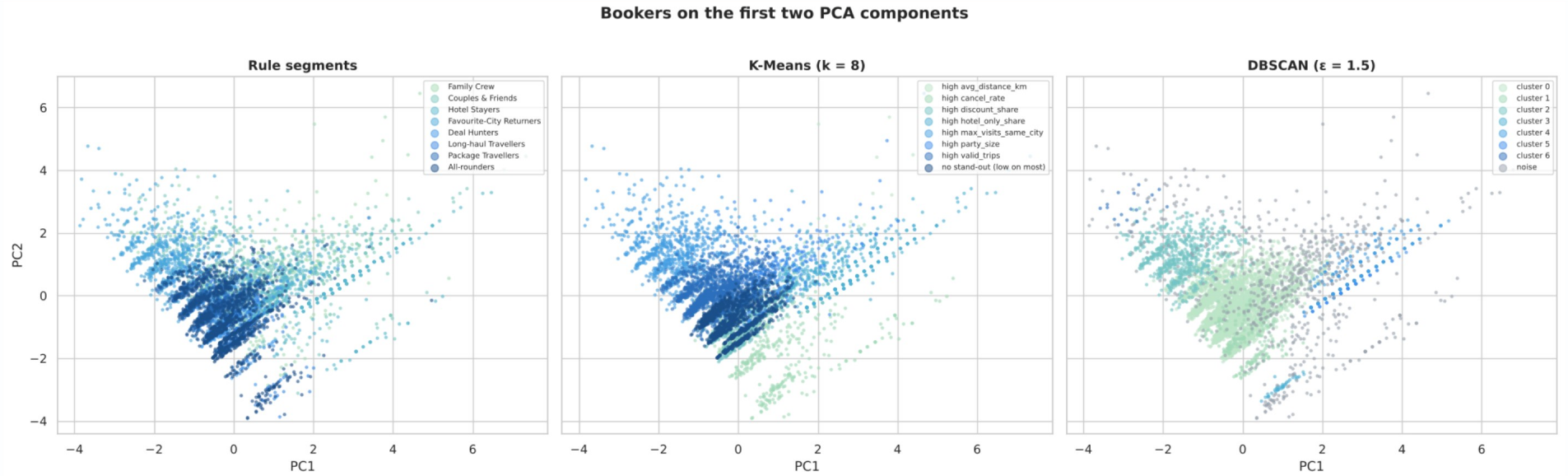
$|r| \leq 0.12$

The perk scores barely overlap. The two hotel perks even reach opposite customers (r = −0.52).

Details: notebook 03, sections 3 and 4.1

BACKUP · UNDER THE HOOD 3/3

Machine learning as a cross-check



K-Means (k = 8)
found 4 of 7 booker groups on its own

DBSCAN
13 % outliers – 97 % of our extreme users

PCA
7 of 9 features needed – no shortcut

Details: notebook 03, sections 5–7 · map = all 5,442 bookers on the first two PCA components